SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – EXPERIMENTS

Course Code:	CSA0620	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	**CO3: Articulation** (BL4, BL5)	**Assessment:**	Assessment Tool 1 – Experiments
Weightage:	40% of CIA	**Total Marks:**	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	**D.HEMASREE**	**Reg. No.:**	**192524114**
Section / Batch:	**2025**	**Date:**	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Write the algorithm and execute the code for the given experiment.
- Follow a well-structured, systematic approach to design and implement an optimal solution.
- Provide insightful analysis with accurate validation, supported by clear documentation including diagrams, code, and results.

Question Type	Focus Area	Questions
Experiments	Design and Code for Divide and Conquer Algorithms: Merge Sort, Quick Sort, Binary Search, and Matrix Multiplication	Q1

ANSWER:

1. **Binary Search is a searching algorithm used to find an element in a sorted array or list.**
2. **It works by comparing the target element with the middle element of the array.**
3. **If the target matches the middle element, the search is completed successfully.**
4. **If the target is smaller, the search continues in the left half of the array.**
5. **If the target is larger, the search continues in the right half of the array.**
6. **This process is repeated until the element is found or the search range becomes empty.**
7. **Binary Search requires the input data to be sorted before searching.**
8. **It reduces the search space by half in every iteration, making it very efficient.**
9. **The algorithm performs fewer comparisons than Linear Search, especially for large datasets.**

10. **The time complexity is O(log n) in the best, average, and worst practical search cases, making it one of the fastest searching techniques for sorted data.**
11. **It is widely used in databases, search engines, dictionaries, phone books, and many real-time applications.**
12. **Binary Search improves search performance, reduces execution time, and efficiently utilizes memory for large collections of data.**

Q1 **Merge Sort** 10 marks

Analyze Merge Sort�s memory utilization by tracking auxiliary space usage during execution. Record memory consumption for different input sizes. Interpret how additional space contributes to stability. Justify trade-offs between time and space complexity.

Your Code

```
import sys

max_space = 0

def merge_sort(arr):
    global max_space
    if len(arr) > 1:
        mid = len(arr) // 2
        left, right = arr[:mid], arr[mid:]
        space_used = sys.getsizeof(left) + sys.getsizeof(right)
        max_space = max(max_space, space_used)

        merge_sort(left)
        merge_sort(right)

        i = j = k = 0
        while i < len(left) and j < len(right):
            arr[k] = left[i] if left[i] <= right[j] else right[j]
            i, j = (i+1, j) if left[i] <= right[j] else (i, j+1)
            k += 1
        arr[k:] = left[i:] + right[j:]

n = int(input())
arr = list(map(int, input().split()))
merge_sort(arr)
print("Sorted array:", arr)
print(f"Input size: {n}, Max Auxiliary Space used: {max_space} bytes")
```

Input

```
7
38 27 43 3 9 82 10
```

Output

```
Sorted array: [3, 9, 10, 27, 38, 43, 82]
Input size: 7, Max Auxiliary Space used: 168 bytes
```

SECTION A – Experiments

Code of Conduct

I _______D.Hemasree(192524114)____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

RUBRICS FOR CALCULATION

Experiments					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Concepts	25%	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30%	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20%	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15%	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10%	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%				
Experimental Design	30%				
Use of Tools	20%				
Analysis of Results	15%				
Documentation	10%				
Total Marks (100):					

Faculty In-charge	**Course Coordinator**	**Head of Department**
Name:	Name:	Name:
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________